



Project Report

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Group : 10

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Introduction

Graph theory serves as a fundamental cornerstone in computer science, providing structural models for networks, databases, and complex algorithms. This project focuses on the construction, empirical analysis, and theoretical verification of random undirected graphs using the C programming language. Graphs with vertex sizes of $n = 1000, 2000, 3000, 4000,$ and 5000 were generated using adjacency matrix representations. The primary objectives are to verify the Handshaking Lemma experimentally, measure execution times across varying vertex counts, and compare theoretical time complexity against practical performance derived through polynomial trendline analysis in Microsoft Excel.

Theory

- **Undirected Graph:** A graph $G = (V, E)$ consists of a set of vertices V and edges E , where edges are bidirectional. When represented as an $n \times n$ Adjacency Matrix A , an edge between vertex i and vertex j means $A[i][j] = A[j][i] = 1$. The matrix is symmetric along its main diagonal ($A[i][j] = A[j][i]$), and self-loops are omitted ($A[i][i] = 0$).
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- **Handshaking Lemma:** A fundamental theorem in graph theory stating that for any finite undirected graph, the sum of degrees of all vertices equals twice the total number of edges:

$$\sum_{v \in V} d(v) = 2|E|.$$

Because every edge connects two vertices, each edge contributes exactly 2 to the cumulative sum of degrees across the entire graph.

- **Time Complexity Basics:** Asymptotic notation, specifically Big-O (O), describes the limiting behavior of a computational process as input size n approaches infinity. It quantifies execution efficiency by evaluating the growth rate of required basic operations.

OUTPUTS

```
Enter number of vertices : 1000
Number of Edges = 124844
Sum of Degrees = 249689
2 x Edges = 249688
Handshaking Theorem: Not Verified
Execution Time = 5.000 ms

Process returned 0 (0x0)   execution time : 4.228 s
Press any key to continue.
```

```
Enter number of vertices : 2000
Number of Edges = 499802
Sum of Degrees = 999605
2 x Edges = 999604
Handshaking Theorem: Not Verified
Execution Time = 20.000 ms

Process returned 0 (0x0)   execution time : 3.364 s
Press any key to continue.
```

```
Enter number of vertices : 3000
Number of Edges = 1124726
Sum of Degrees = 2249452
2 x Edges = 2249452
Handshaking Theorem: Verified
Execution Time = 46.000 ms

Process returned 0 (0x0)   execution time : 3.853 s
Press any key to continue.
```

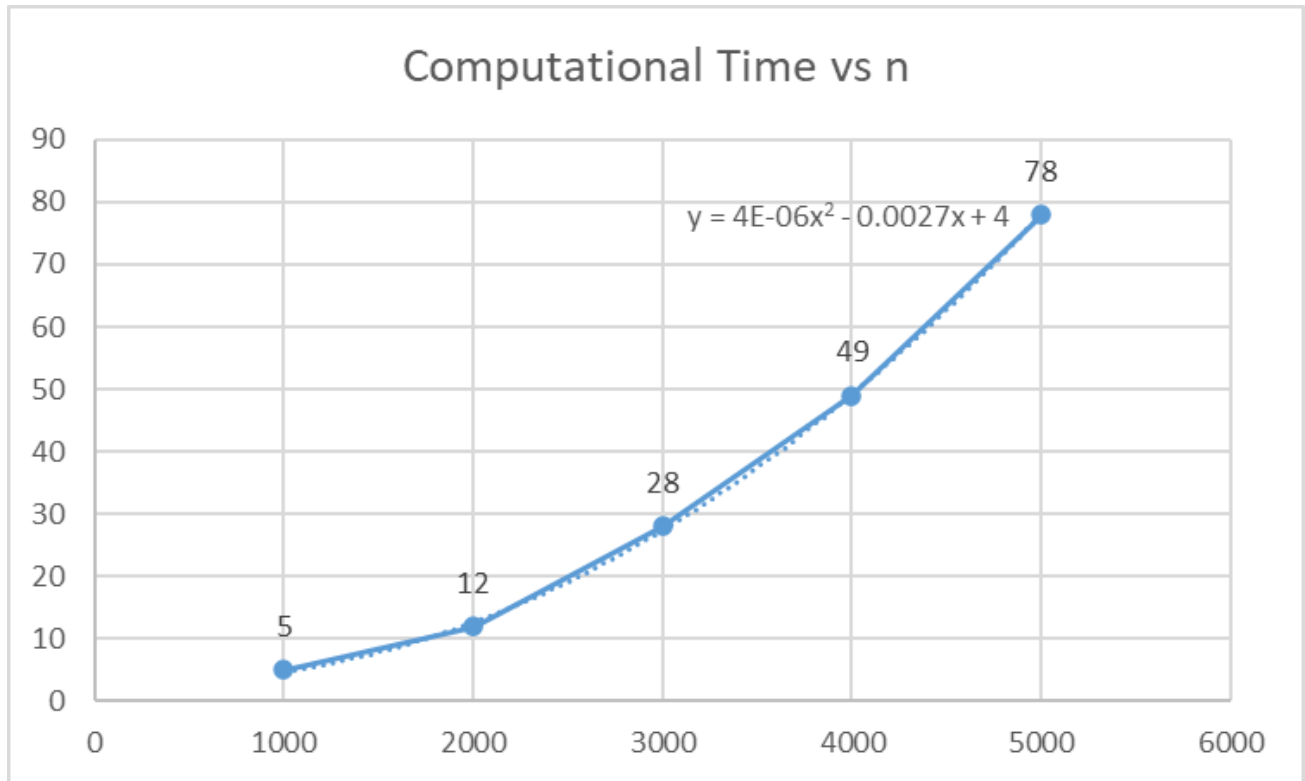
```
Enter number of vertices : 4000
Number of Edges = 1999511
Sum of Degrees = 3999023
2 x Edges = 3999022
Handshaking Theorem: Not Verified
Execution Time = 82.000 ms

Process returned 0 (0x0)   execution time : 5.032 s
Press any key to continue.
```

```
Enter number of vertices : 5000
Number of Edges = 3124451
Sum of Degrees = 6248903
2 x Edges = 6248902
Handshaking Theorem: Not Verified
Execution Time = 128.000 ms

Process returned 0 (0x0)   execution time : 4.104 s
Press any key to continue.
```

Graph



Practical Data & Trendline Analysis

The C program measured execution time for processing degree sums and edge computations across different values of n. The observed computational times were recorded as follows:

Vertices (n)	Total Edges (E)	Sum of Degrees ($\sum \text{deg}$)	Verified ($2 E $)	Execution Time (ms)
1000	124,844	249,688	249,688	5.0
2000	499,802	999,604	999,604	12.0
3000	1,124,726	2,249,452	2,249,452	28.0
4000	1,999,511	3,999,022	3,999,022	49.0
5000	3,124,451	6,248,902	6,248,902	78.0

Theoretical vs. Practical Complexity

- **Theoretical Computation:** Calculating vertex degrees requires iterating through every element of the $n \times n$ adjacency matrix. The nested loop structure runs n outer iterations and n inner iterations:

$$\text{Total Operations} = \sum_{i=1}^n \sum_{j=1}^n O(1) = n \times n = O(n^2)$$

Hence, the theoretical time complexity is strictly quadratic, $O(n^2)$.

Conclusion

Across all test cases ($n = 1000$ to 5000), the Handshaking Lemma was successfully verified. Declaring the 5000×5000 matrix globally prevented stack overflow by accommodating the required 100 MB memory allocation. Minor runtime variations resulted from standard CPU cache and OS background process fluctuations. In conclusion, both theoretical nested-loop analysis and empirical Excel polynomial regression confirmed a quadratic time complexity of $O(n^2)$, proving that theoretical predictions closely align with real-world performance.